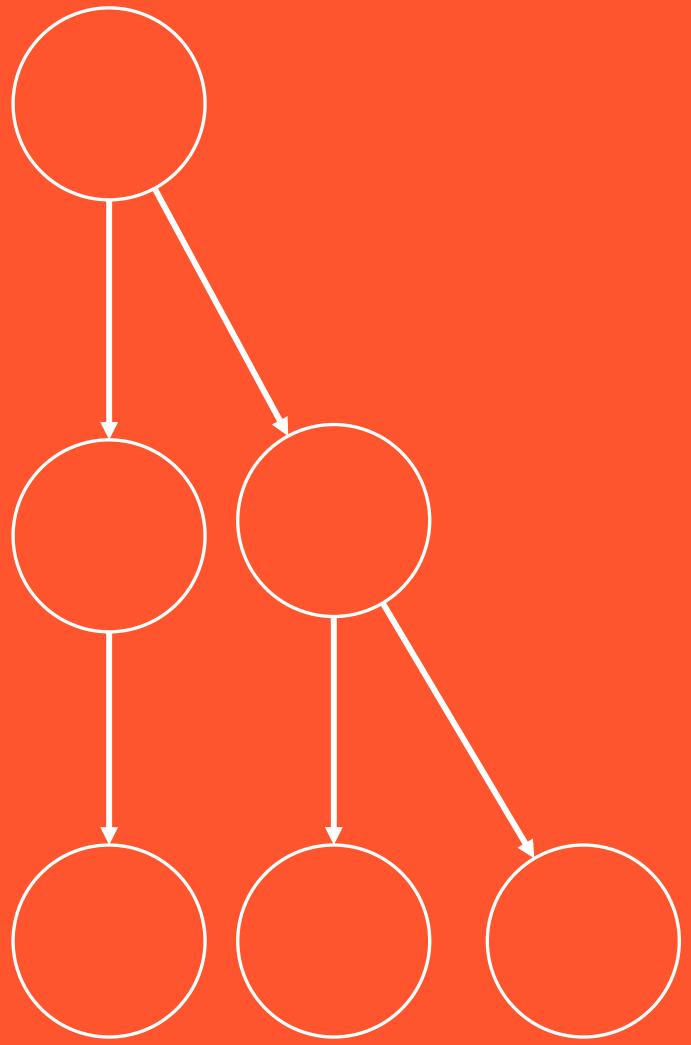


BCOG 100

Neural Circuits, Valence, and Affect

Lecture 4.1 — The First Nervous Systems and Their Value

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

As Life Got More Complex, New Options Opened Up

A sea anemone, a moon jelly, a nematode worm. Three animals alive right now, and three different answers to what a body can do about the world. One of those answers is **steering**.







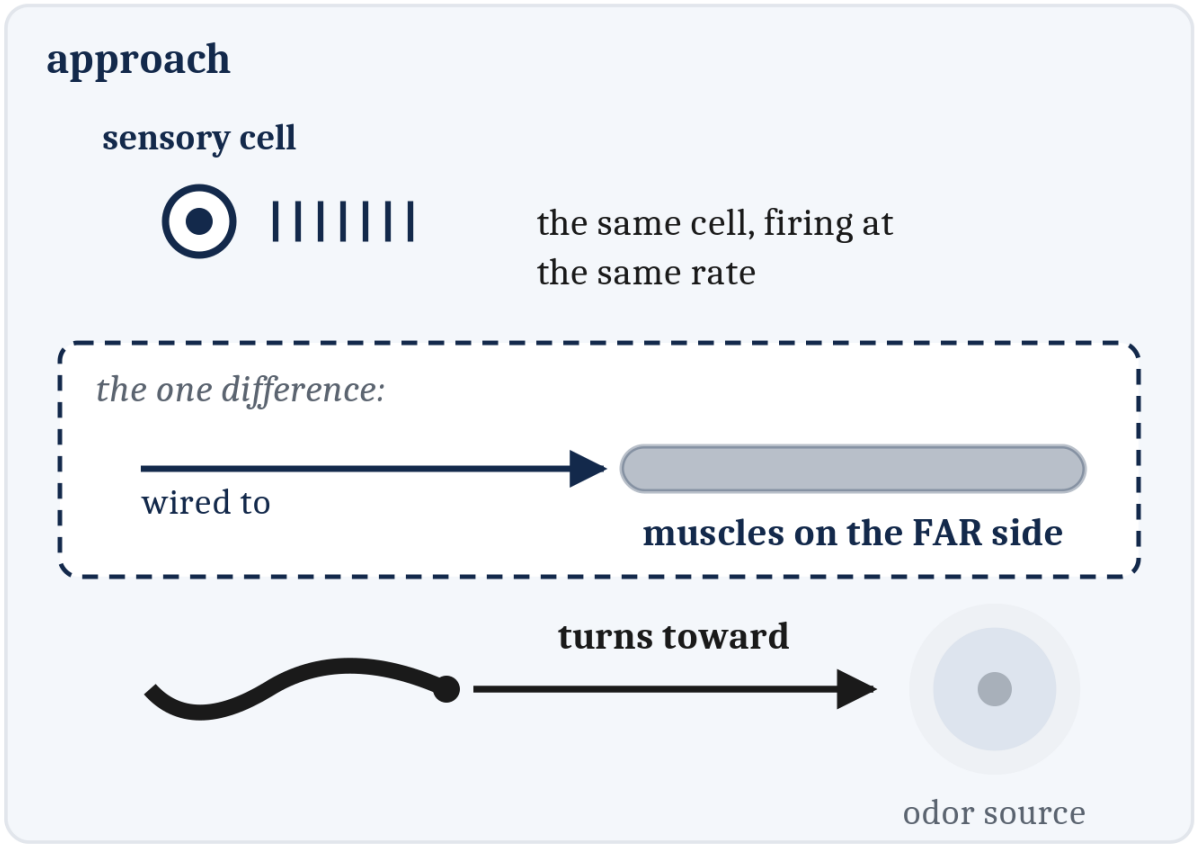
One Capacity, Three Questions

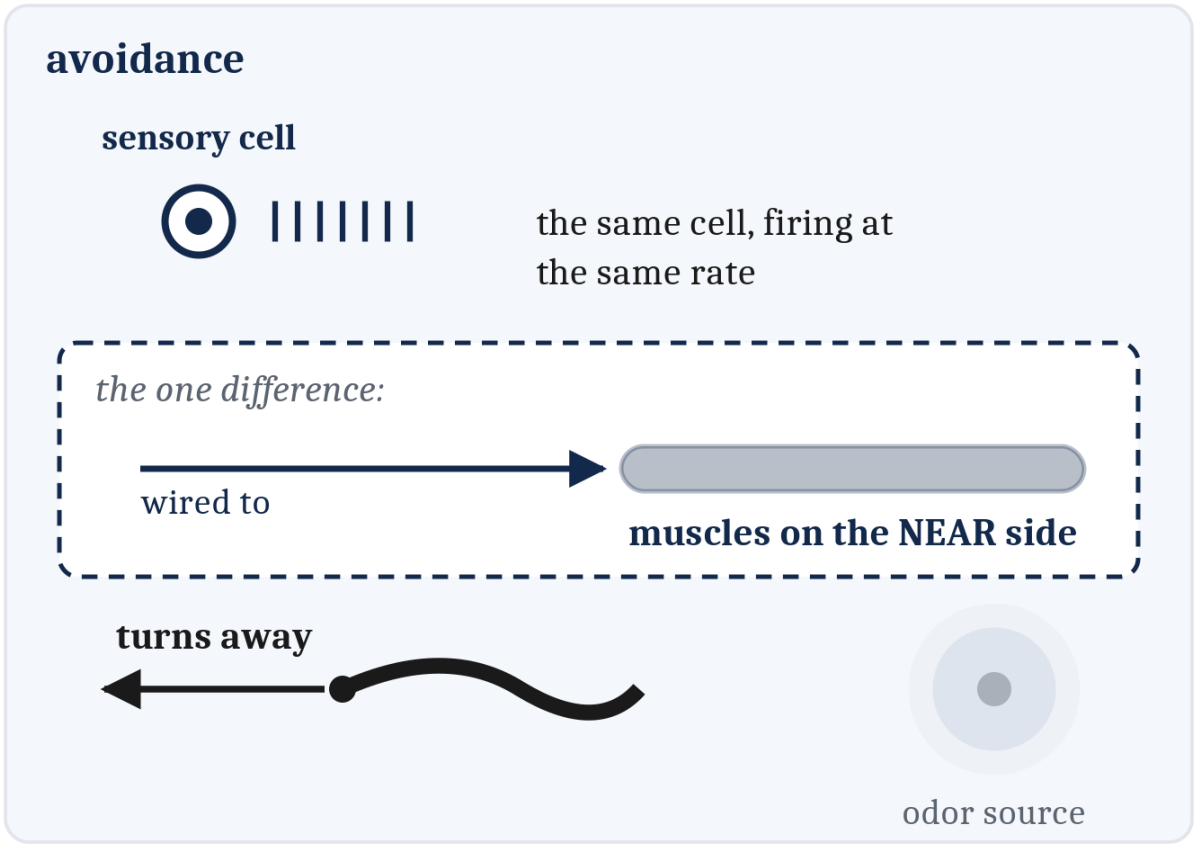
The second module built this way, and the rhythm now holds to the end of the course. One capacity, three sections in the reading, three meetings, one per level.

Level	The question it asks	Discipline	This module's example: the steering circuit
Computational	What problem is being solved, and why does solving it matter?	Evolution and ecology	Steering — aiming a body at something, and away from something else
Algorithmic	What is represented, and by what procedure?	Cognitive psychology and AI	Weights, thresholds, recurring wiring patterns, and the valence–arousal pair
Implementational	By what physical means, and at what cost?	Neuroscience	Nerve nets condensing into ganglia, a worm whose wiring diagram is complete, and neuromodulators released into the tissue

The Same Cell, the Opposite Behavior

A rate is only a rate. It becomes *approach* because of what the cell is wired to — and **the identical cell, wired to the opposite muscles, produces avoidance.**

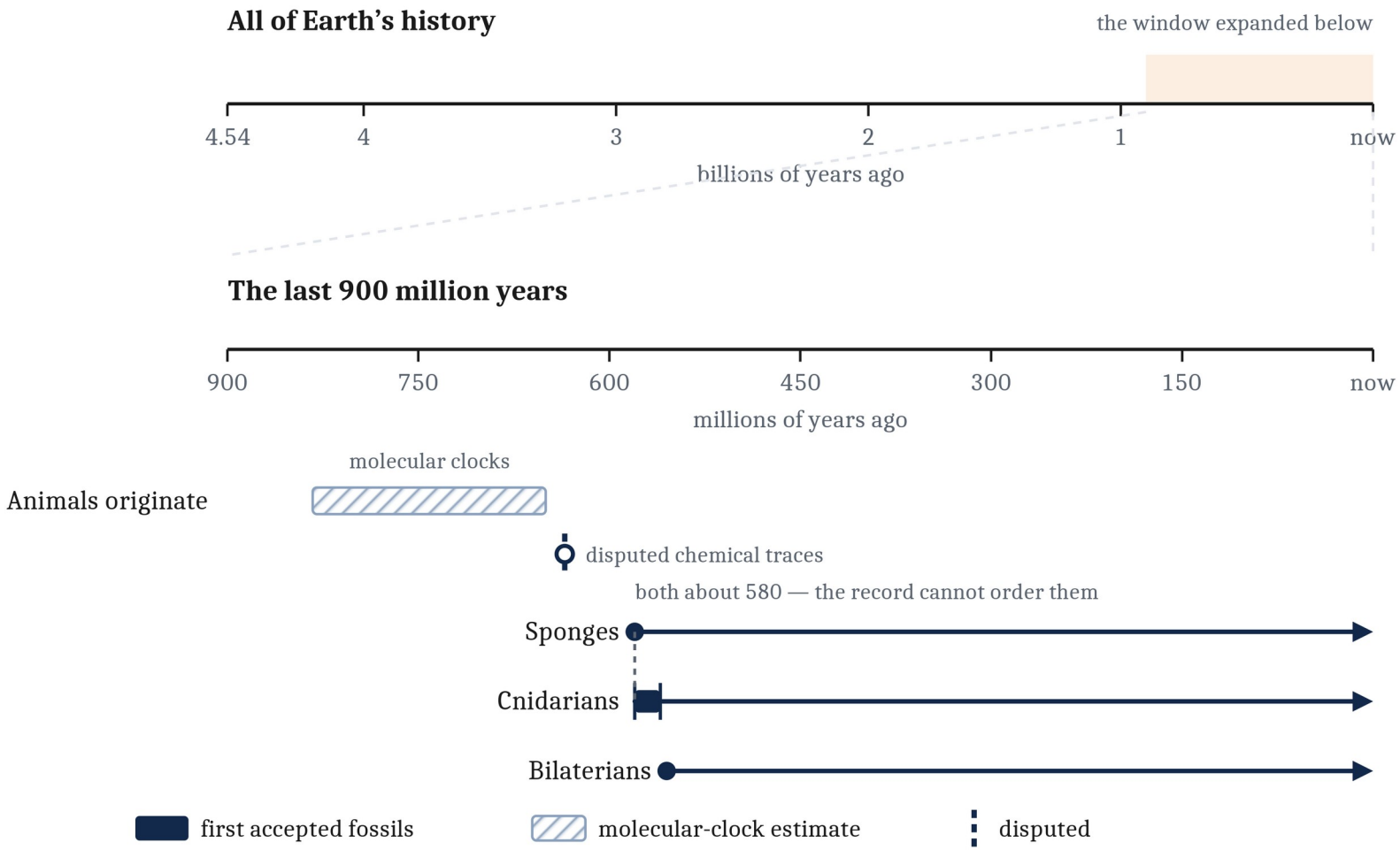




The cell is identical. The rate is identical. Only the wiring differs.

What a Computational Question Asks, and the Warning That Comes With It

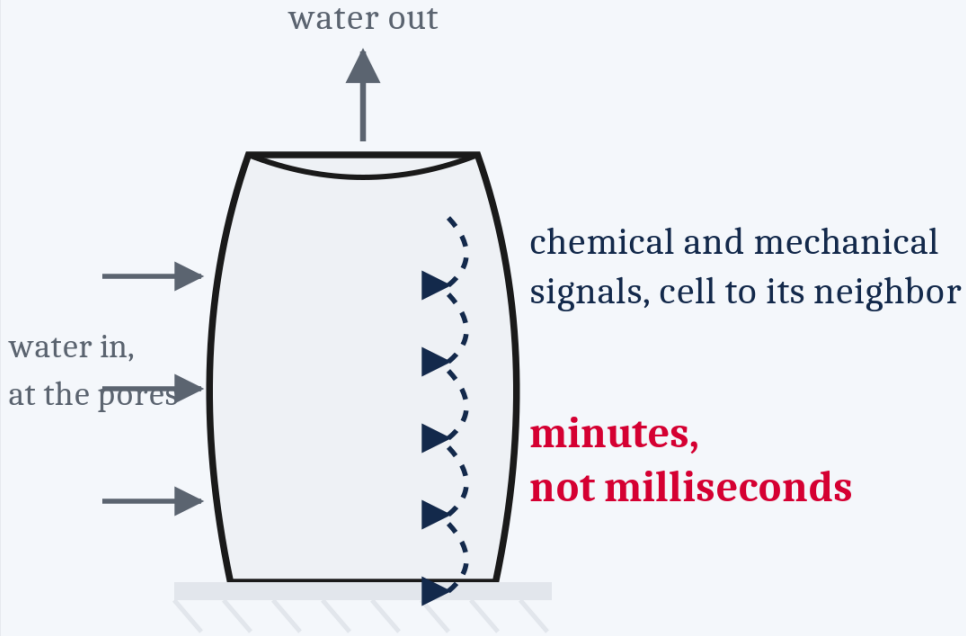
The computational question asks **what problem is being solved, and why solving it was worth what it cost.** Walking forward through the animal tree reads very easily as walking *up* it. It is not one.



Bodies, and Behavior, Came First

A sponge has no neuron, no synapse, no nerve, and no ordinary muscle — and it behaves anyway. **The step to a nervous system is the arrangement, not the invention of a new chemistry.**

a whole-body contraction, and no neurons



coordinated across the whole animal, and fast enough for a filter feeder

and yet, what a sponge already has

no neuron. no synapse. no nerve.



genes for synapse proteins



genes for receptors



genes for the ion channels of an action potential

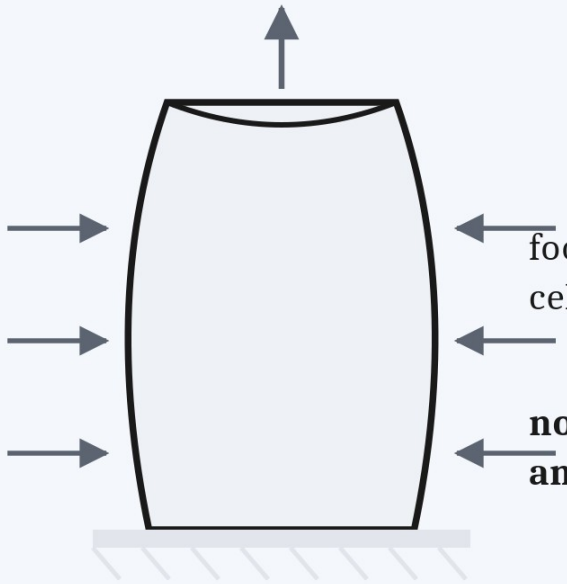
all of it lying loose, and nothing wired to anything

The parts were lying around. What had not happened was the arrangement.

A Mouth to Operate

Filtering asks for no coordination beyond keeping the water moving. Catching does. A **nerve net** — neurons spread through the body wall, each connected to its neighbors — beat the diffusion limit.

filtering

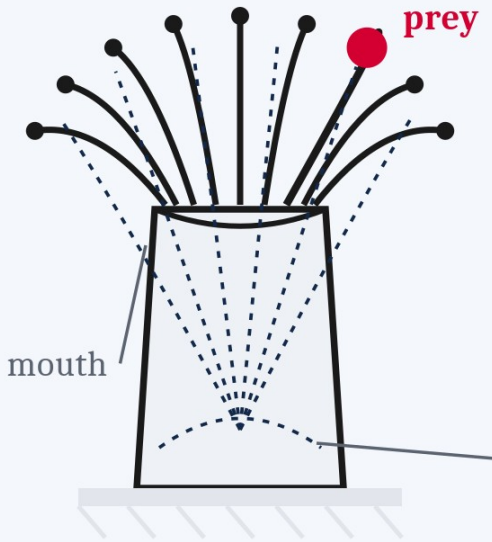


food is caught by single cells, one particle at a time

nothing has to be caught, and nothing swallowed

no coordination beyond keeping the water moving

catching



1. **strike**
2. **hold**
3. **carry**
4. **admit**
5. **close**

nerve net
spread through the animal, not gathered anywhere

Five things, in order, in different parts of the body, before the prey gets away.

The Bell Contracts as One Thing

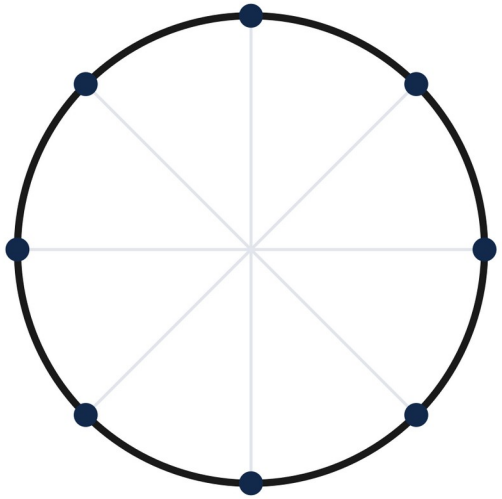
A wave crossing the net reaches every part of the ring within a few thousandths of a second. **Having a nervous system is not the same as having a brain, and neither is the same as making a decision.**



Coordination in Time, Without Direction in Space

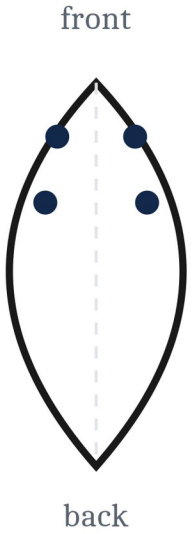
Radial symmetry is a body plan built around a central axis and much the same all the way around, so no part of it ever arrives anywhere first — and therefore there is no privileged place for sensors to gather.

Two body plans, from above



Radial

much the same all the way around;
no part arrives anywhere first



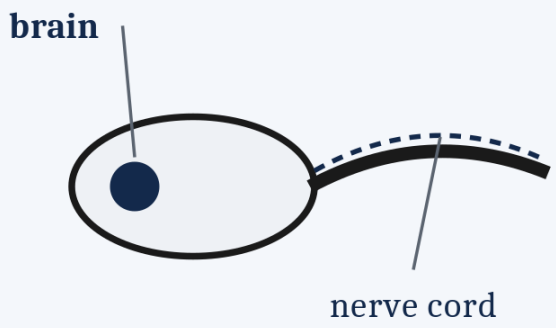
Bilateral

a left and a right that mirror,
and so a front where sensors gather

A Front — and What Having One Costs

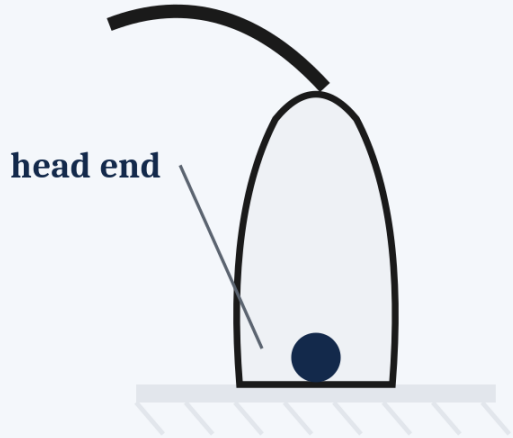
Bilateral symmetry gives an animal a front, and the end that arrives first is the useful place for sensors and for the neurons they report to. **Steering** is moving forward, turning, and letting what the front end detects decide which way the turn goes.

a swimming larva



it steers, looking for somewhere to settle

attaching, head first



head first -- and then it will never move again

the settled adult



no tail. no brain -- just enough to run the siphons and the gut

An animal that stops steering does not go on paying for the machinery that steers.

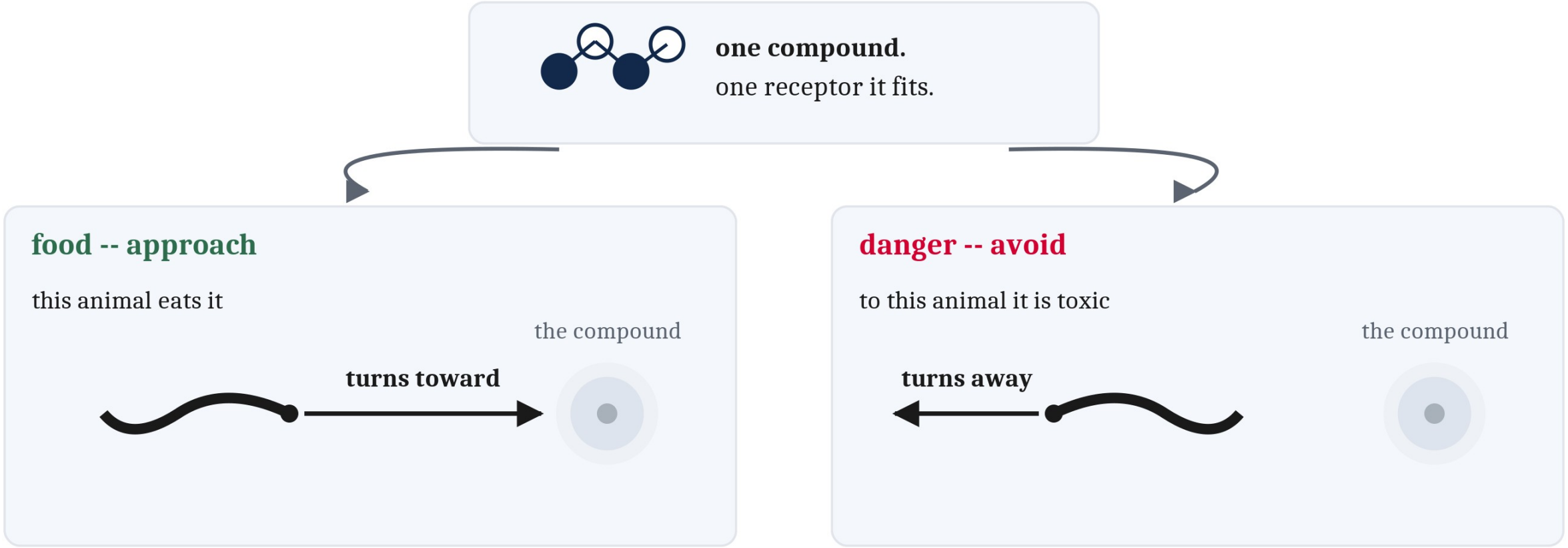
What Steering Demands

Steering is not free. Four demands, and the rest of the module is organized around them — we state them here, build the arrangements in the next lecture, and find them in tissue in the one after.

Demand	What steering requires	Built in 4.2 as	Found in 4.3 in
Evaluate	Sort what is sensed into good, bad, and neither — no direction can be preferred without a reason to prefer it	The labeled line	The worm's sensory neurons
Combine	Weigh several cues at once, both by how strong each one is and by which combination of them is present	Weights, thresholds, and logic units	The worm's steering circuit
Sharpen and commit	Turn a near-tie into a clear difference, then collapse many competing votes into exactly one action	Lateral and mutual inhibition	The two ends of the circuit
Modulate	Let the animal's own state change the verdict, and let a state outlast the event that caused it	Feedback loops	The neuromodulators

Evaluate: Where Good and Bad Come From

Valence is the tag a nervous system attaches to what it senses, marking a thing as worth moving toward or away from. There is no goodness in the molecule — and none in the sensory cell that detects it either.



The difference is not in the molecule. Both verdicts are correct for the animal holding them.

Combine: Two Ways Cues Have to Be Put Together

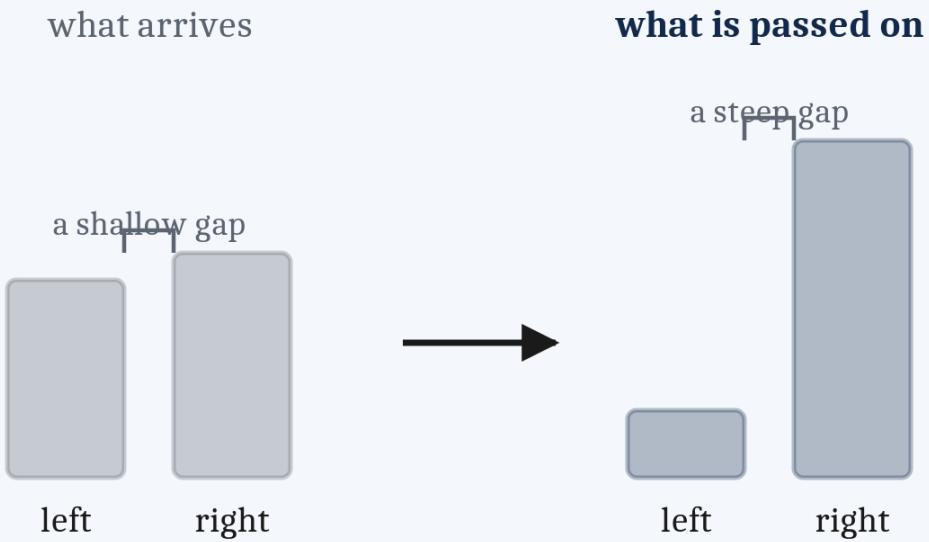
Combination has a **quantitative** face — graded strengths weighed against each other — and a **relational** one, where what matters is which combination is present. A **truth table** states the second exactly; the mapping it states is the circuit's **target function**, and a piece of nervous system that computes one of these relations is a **logic gate**.

Cue A	Cue B	A AND B	A OR B	A AND NOT B
absent	absent	do not act	do not act	do not act
absent	present	do not act	act	do not act
present	absent	do not act	act	act
present	present	act	act	do not act

Sharpen, and Then Commit

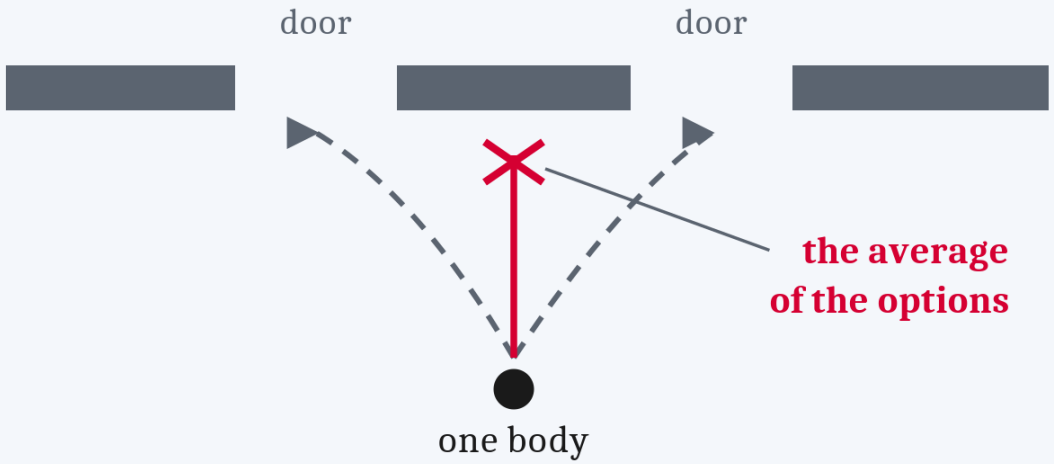
A nervous system that reported a near-tie faithfully would hand the muscles a near-tie. Sharpening takes a shallow gap in and produces a steep one out — and then committing follows from a fact about bodies rather than a fact about information.

sharpen



the marginally better direction becomes plainly the better one

commit



not a blend. not half a turn each way.

A system that averages its options is a system that walks between two doors into the wall.

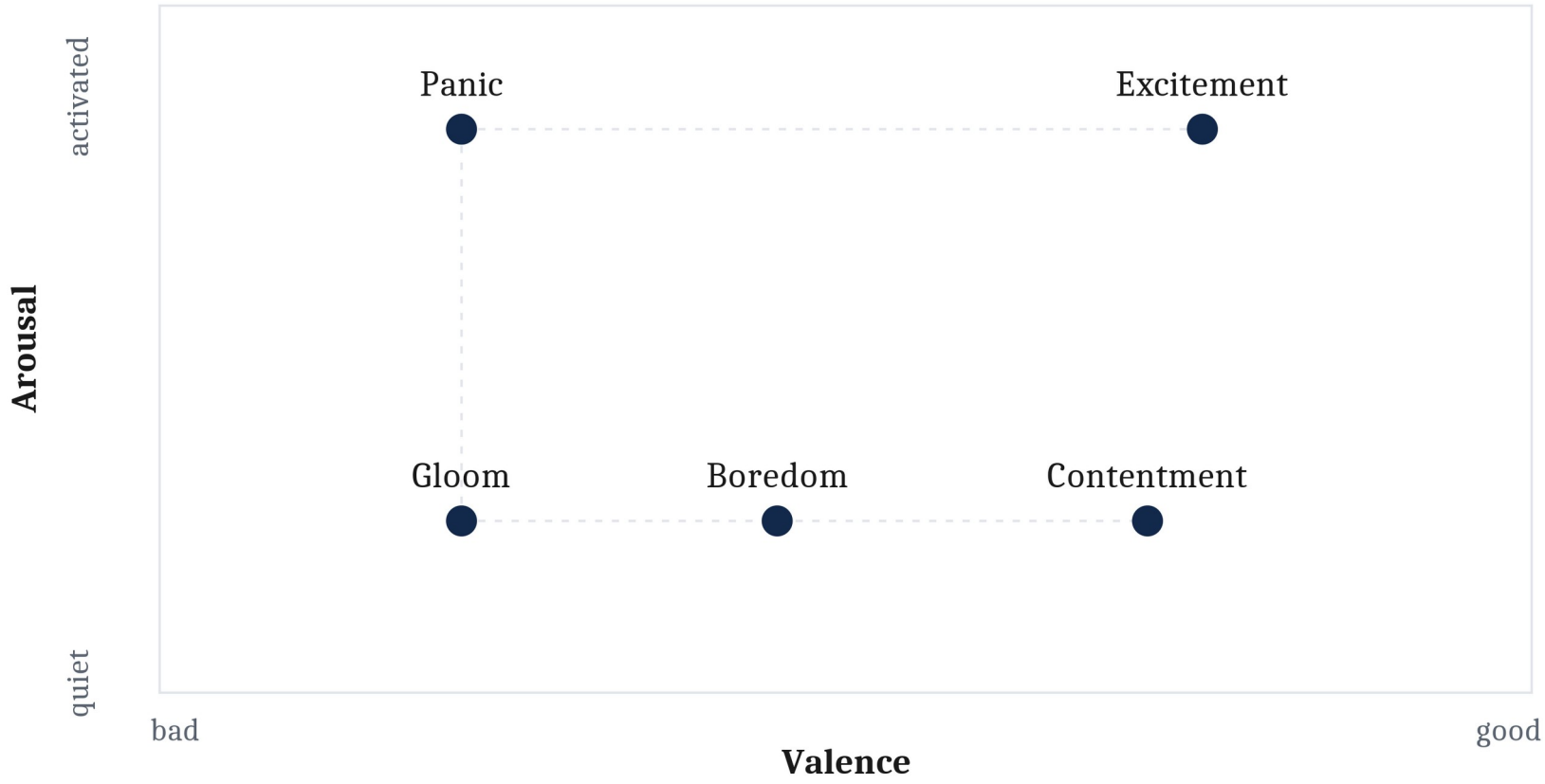
Modulate: The Same Animal, the Same Molecule, the Opposite Behavior

An **internal state** is the animal's own condition — hunger, injury, fatigue — and the fourth demand is that this condition reach into the evaluation and change the verdict.

The worm's condition	The cue	The verdict
Hungry	Food odor across a copper strip	Worth crossing
Just eaten	The same odor, the same copper	Little to gain
Recently fed	Carbon dioxide	Avoid it
Some hours without food	The same carbon dioxide	Avoid it far less
Under some conditions	The same carbon dioxide	Drawn toward it

Affect: A State That Outlasts Its Cause

Affect is an internal state that outlasts its cause and biases behavior over a stretch of time. **Arousal** is how activated an animal is, and with valence it gives a two-dimensional space.



Where Do These Sit?

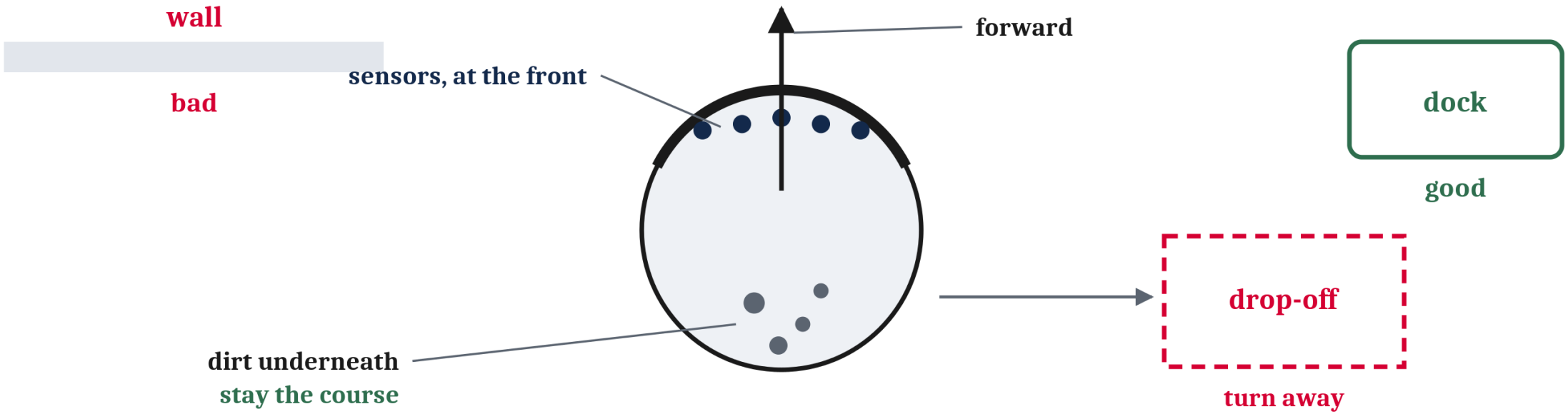
Place each state in the valence–arousal space. Where does the room disagree?

- Waiting for a grade to post
- The last mile of a long run
- Scrolling at one in the morning
- Homesick
- Ten minutes into a lecture on something everyone already knows

Then: which of these outlasts its cause the longest?

A Machine That Steers

A robot vacuum meets all four demands, and it is not trying to be an animal. The four demands belong to **the problem** rather than to biology.



1. evaluate	a wall is bad and the dock is good -- and neither fact is in the object
2. combine	dirt underneath, against a drop-off: settled together, not obeyed in turn
3. sharpen and commit	one path at a junction, never a blend of two
4. modulate	the dock is good only when the battery is low, and a dirty patch keeps being worked

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